

LAGHOUAT, Algeria

WMO: 605450

Lat: 33.764N

Lon: 2.927E

Elev: 765

StdP: 92.47

Time Zone: 1.00 (E01)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.6	-7.8	2.1	12.1	-6.1	2.5	11.2	16.9	11.4	14.4	10.0	0.5	240	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	41.2	20.1	40.1	19.9	38.9	19.6	22.7	36.0	21.7	35.5	20.9	34.7	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	15.1	27.6	17.8	14.0	26.6	16.8	13.1	25.9	70.8	36.0	66.8	35.3	64.0	34.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.3	11.5	10.1	DB	-3.4	43.0	1.4	0.7	-4.4	43.5	-5.2	43.9	-6.0	44.3	-7.0	44.8	
			WB	-4.3	24.4	1.2	1.6	-5.1	25.6	-5.9	26.5	-6.6	27.3	-7.4	28.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	8.5	9.9	13.7	17.9	22.7	28.0	32.1	30.7	25.5	20.0	13.0	9.2
	DBStd	8.74	2.57	3.25	3.33	3.72	3.66	3.32	2.16	2.41	3.31	3.24	2.98	2.69
	HDD10.0	158	61	37	7	0	0	0	0	0	0	0	6	47
	HDD18.3	1213	305	237	146	53	7	0	0	0	1	19	162	283
	CDD10.0	3560	13	34	123	236	393	541	685	641	466	309	97	22
	CDD18.3	1574	0	1	3	39	141	291	427	383	217	71	3	0
	CDH23.3	22233	0	8	89	499	1721	4112	6848	5696	2507	718	34	1
	CDH26.7	13124	0	1	12	153	804	2448	4588	3596	1279	239	4	0
(14) Wind	WSAvg	3.9	3.6	4.6	4.7	5.0	4.8	4.5	3.9	3.6	3.4	2.8	3.4	3.1
(15) Precipitation	PrecAvg	167	13	9	12	16	15	10	6	12	22	18	15	16
	PrecMax	287	64	42	51	87	53	36	30	79	79	81	95	70
	PrecMin	88	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	62	16	11	14	19	15	11	7	16	19	21	22	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	24.2	28.2	32.9	37.0	40.9	43.0	42.1	38.5	34.1	26.7	21.8
		MCWB	10.9	11.7	13.5	15.3	17.6	20.1	19.8	20.2	18.7	18.3	14.4	12.2
	2%	DB	18.8	21.1	25.9	30.2	34.9	39.2	42.0	40.8	36.2	31.5	23.8	19.0
		MCWB	10.1	11.0	12.6	14.7	17.4	19.2	20.1	19.9	19.1	17.0	13.6	11.4
	5%	DB	16.9	18.9	23.8	28.2	33.0	37.9	40.9	39.2	35.0	29.2	21.5	17.0
		MCWB	9.4	10.1	12.1	14.2	17.0	19.0	20.3	19.7	18.9	16.6	12.7	10.3
	10%	DB	15.0	16.9	21.9	26.2	31.1	36.1	39.8	38.0	33.1	27.2	19.8	15.0
		MCWB	8.7	9.1	11.2	13.7	16.2	18.8	20.1	19.7	18.6	16.2	12.2	9.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.4	15.1	17.7	20.8	24.1	24.7	23.7	22.2	20.1	16.6	13.2
		MCDB	18.8	22.2	24.5	28.5	31.0	35.8	38.0	36.5	33.5	28.0	23.8	18.5
	2%	WB	10.9	11.8	13.7	16.0	19.1	22.0	22.8	22.3	21.0	18.7	15.0	11.8
		MCDB	17.1	19.2	23.5	27.1	30.8	34.8	37.0	35.8	31.6	26.6	21.0	17.3
	5%	WB	9.8	10.6	12.8	15.0	18.1	20.5	21.8	21.3	20.2	17.8	13.6	10.8
		MCDB	15.9	17.4	22.0	25.0	29.8	34.0	36.8	34.7	30.4	26.0	19.7	16.1
	10%	WB	8.9	9.5	11.7	14.2	17.1	19.5	20.9	20.7	19.6	17.1	12.6	9.8
		MCDB	14.5	15.9	20.2	24.4	28.6	33.4	36.5	33.9	29.6	24.8	18.4	14.3
(35) Mean Daily Temperature Range	MDBR		12.3	12.1	13.1	13.3	13.7	14.1	14.6	14.1	13.1	12.8	11.9	11.5
	5% DB	MCDBR	15.9	15.8	16.7	16.1	15.8	15.5	15.7	15.3	15.0	15.3	14.8	15.0
		MCWBR	9.1	8.0	7.6	6.6	6.0	5.6	5.2	5.3	5.1	5.9	7.0	8.4
	5% WB	MCDBR	14.6	14.6	14.9	14.3	14.3	14.3	14.6	14.2	13.5	12.8	12.8	13.0
		MCWBR	8.8	7.8	7.2	6.5	6.2	6.4	6.0	5.4	5.4	5.8	6.7	7.8
(40) Clear-Sky Solar Irradiance	taub		0.281	0.311	0.389	0.426	0.478	0.510	0.570	0.533	0.487	0.417	0.335	0.294
	taud		2.444	2.357	2.105	2.023	1.901	1.829	1.706	1.793	1.906	2.077	2.325	2.422
	Ebn at Noon		939	938	881	862	819	790	740	760	774	802	858	899
	Edh at Noon		88	108	151	172	196	211	237	214	182	141	99	86
(44) All-Sky Solar Radiation	RadAvg		3.35	4.29	5.56	6.64	7.34	7.90	7.72	6.95	5.68	4.61	3.53	3.03
	RadStd		0.21	0.25	0.27	0.29	0.33	0.23	0.20	0.23	0.27	0.31	0.17	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.31	N/A	-1.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page